Package ‘cdmetapopR’

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Title Utility functions for CDMetaPOP outputs

Version 0.0.1

Description This package provides a set of helper functions to analyze and visualize the output files of CDMetaPOP

Imports ggplot2,
reshape2,
tidyr,
dplyr

License GPL (>= 2)

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Suggests testthat (>= 3.0.0)

Config/testthat/edition 3

Contents

`age_plus_one_plot` *Function to extract population sizes of age +1 individuals*

Description

This function produces a plot with population sizes of age 1+ individuals. It is based on the N_Initial_class column of the CDMetaPOP output file summary_classAllTime.

Usage

```
age_plus_one_plot(data)
```

Arguments

`dataframe`	summary_classAllTime.csv dataframe

Value

ggplot

`age_structure_proportions` *Determine the proportions of the age structure of the population*

Description

This function summarizes the proportions of the age structure in the simulated population. It is calculated counting the values in the column 'age' of the chosen 'ind' generation.

Usage

```
age_structure_proportions(path = mypath, runs = 1, gen = 49, species = 0)
```

Arguments

`path`	path to the simulation folder
`runs`	integer referring to the number of montecarlo simulations output files to examine. It refers to the mcruns option in the RunVars CDMetaPOP input file. Right now it defaults to 1, which would be only one montecarlo run.
`gen`	Simulation run time (generation or year). Refers to runtime in the RunVars of CDMetaPOP input file. Currently defaults to 49.
`species`	If the simulation includes more species... The code for this needs to be adjusted

Value

a dataframe with a column for each montecarlo run (MC) with the proportion of the population for each given age (in rows) and the last column with the average value across all montecarlo runs at any given age.

Examples

```
define  your path to the output_test folder: mypath = "~/UM/CDMetaPop/EBT/EBT_Inputs/data/output_test17284877
```

`alleles_by_year` *Function extracts and plots allele count overtime*

Description

This function plots the total number of unique alleles for every year of the simulation. It is based on the Allele column of the CDMetaPOP output file summary_popAllTime

Usage

```
alleles_by_year(data, n = 5)
```

Arguments

`n`	An integer that specifies the time intervals of the time series. It defaults to 5 years.
`dataframe`	summary_popAllTime.csv

Value

a ggplot object representing the overtime observed and expected heterozygosities values in the simulation.

`births` *Create a plot with the proportion of young of the year. Note here that this proportion includes only age 0 progeny that survived egg mortality.*

Description

This function plots the proportion of progeny over the total, after egg deaths are taken into account

Usage

```
births(file_path)
```

Arguments

`x`	Specify the path to the summary_popAllTime.csv dataframe

Value

a ggplot object representing the size of progeny surviving egg deaths every year

cdmetapop_to_gene — *Convert CDMetaPOP data to GENEPOP or GENALEX format*

Description

This function reads a CDMetaPOP CSV file, processes the genetic loci, and converts the data into GENEPOP or GENALEX formats.

Usage

```
cdmetapop_to_gene(path, format = "genepop")
```

Arguments

path	Character string; the file path to the CDMetaPOP CSV file.
format	A character string indicating the desired output format: "genepop" or "genalex". Genepop is default.

Value

The function writes the GENEPOP or GENALEX formatted data into the working directory.

Examples

```
## Not run:
# Convert a CDMetaPOP file to GENEPOP format
cdmetapop_to_gene(
  path = "your_path/ind1.csv",
  format = "genepop"
)

# Convert a CDMetaPOP file to GENALEX format
cdmetapop_to_gene(
  path = "your_path/ind1.csv",
  format = "genalex"
)

## End(Not run)
```

create_cdmat — *Create cost distance matrix for CDMetaPOP simulation*

Description

Takes input coordinates and produces a symmetrical cost distance matrix with options for Euclidean distance, equal distance, or least cost paths based on a provided resistance surface

Usage

```
create_cdmat(
  coords,
  method = c("euclidean", "equal", "lcp"),
  resistance = NULL
)
```

Arguments

`coords`	A set of spatial coordinates in the form of a 2-column data frame or matrix
`method`	Method for creating the cost distance matrix, must be "Euclidean", "Equal", or "LCP"
`resistance`	Raster object representing resistance surface for generating cost distance matrix

Details

Hello

Value

An NxN matrix of cost distances among patch locations

Examples

```
x <- c(1,2,3,4,5,6)
y <- c(1,2,3,4,5,6)
test <- data.frame(x=x, y=y)
create_cdmat(coords=test)

x <- runif(6, min=0, max=10)
y <- runif(6, min=0, max=10)
coords <- data.frame(x=x, y=y)
r <- terra::rast(nrows=10, ncols=10, xmin = 0, xmax = 10, ymin = 0, ymax = 10)
terra::values(r) <- 1
create_cdmat(coords=coords, method="LCP", resistance=r)
```

`dispersal` *Counts number of individuals that move each year by reading in each ind file*

Description

Counts number of individuals that move each year by reading in each ind file

Usage

```
dispersal(
  path = path,
  run = 0,
  batch = 0,
  mc = 0,
  gen = 49,
```

```
  species = 0,
  ind = TRUE,
  plot = FALSE
)
```

Arguments

path	Character. The path to the directory containing the output files.
run	Integer. The run index. A run is each row of the RunVars file - Defaults to 0.
batch	Integer. The batch index. See RunVars PreProcess Run parameters in the CD-MetaPOP user manual.
mc	Integer. The Monte Carlo index. It refers to the number of Monte Carlo simulations output files to examine. It refers to the mcruns option in the RunVars CDMetaPOP input file. It defaults to 0, which is one run.
gen	Integer. The number of generations. Simulation run time (generation or year). Refers to runtime in the RunVars of CDMetaPOP input file. It defaults to 49.
species	Integer. The species index. If the simulation includes multiple species. It defaults to 0, meaning one species only.
ind	Logical. Whether to use ind files ('TRUE') or sample files ('FALSE').
plot	Logical. If TRUE, a histogram of the proportion of movers across the ind file defined by the chosen parameters will be displayed. Defaults to FALSE. NOTE: The histogram of movers only reflects the counts for the given combination of parameters (not across all iterations).

Details

This function was partially developed with assistance from ChatGPT, an AI model by OpenAI.

Value

A data frame with proportions of movers or a plot of the count of movers per year. The data returned is a dataframe that lists the proportion of dispersors for each year and for each Monte Carlo simulation.

Examples

```
# Example: using the example directory provided in the package
my_output <- system.file("extdata/output_test1732137064/", package = "cdmetapop_R")
# Alternatively use a custom path:
my_output <- "/your_path/cdmetapop_R_package/inst/extdata/output_test1732137064/"

# Example: Run the function to output a dataframe for all the generations and replicas specified:
 dispersal (path = my_output, run = 1, batch = 0, mc = 2, gen = 9, species = 0, ind = FALSE, plot = FALSE)

# Example: Generating a barplot for the example directory
dispersal(path = my_output, run = 0, batch = 0, mc = 0, gen = 9, species = 1, ind = FALSE, plot = TRUE)
```

extinct_patch — *Check which patch(es) went extinct (no more individuals left) at a particular year*

Description

This function produces a vector with the number of patches that suffered extinction at a given year. It is based off the N_Initial column of the CDMetaPOP output file summary_popAllTime.

Usage

```
extinct_patch(data, year = 100)
```

Arguments

`year`	value referring to the year to select. It defaults to 100
`dataframe`	summary_popAllTime.csv dataframe

Value

integer vector

hets_plot — *Create heterozygosity plot*

Description

This function plots heterozygosity overtime. It is based on the Ho and He columns of the CD-MetaPOP output file summary_popAllTime

Usage

```
hets_plot(data)
```

Arguments

`x`	dataframe summary_popAllTime.csv

Value

a ggplot object representing the overtime observed and expected heterozygosities values in the simulation.

myy_progeny_plot *Plot the proportion of exclusive Myy progeny over the total*

Description

This function plots the ratio of the Myy progeny overtime

Usage

```
myy_progeny_plot(file_path)
```

Arguments

`x`	Specify the path to the summary_popAllTime.csv dataframe

Value

a ggplot object representing the overtime progeny size of YYMales

pop_count_plot *Create abundance by year plot*

Description

This function plots population dynamics overtime showing a count of the individuals over the years of the simulation. It is based off the N_Initial column of the CDMetaPOP output file summary_popAllTime

Usage

```
pop_count_plot(data)
```

Arguments

`dataframe`	Import the summary_popAllTime.csv dataframe

Value

a ggplot object representing the overtime general population size `pop_mature_plot` *Create abundance by year plot showing the numbers of mature individuals of all sexes*

Description

This function plots population dynamics overtime showing a count of the mature individuals over the years of the simulation separated by their sex. It is based off the N_Mature columns of the CDMetaPOP output file summary_popAllTime

Usage

```
pop_mature_plot(file_path)
```

Arguments

`x`	Specify the path to the summary_popAllTime.csv dataframe

Value

a ggplot object representing the overtime population size of sexually mature individuals

`pop_patch_count` *Create abundance by patch plot*

Description

This function plots the population abundance per Patch. It is based off the N_Initial column of the CDMetaPOP output file summary_popAllTime.

Usage

```
pop_patch_count(data, years = c(1, 25, 50, 100))
```

Arguments

`years`	vector of values referring to the years to select. It defaults to c(1,25,50,100)
`dataframe`	summary_popAllTime.csv dataframe

Value

a ggplot boxplot showing the population numbers of different patches pop_sex_plot *Create abundance by year plots for the different sex - Male/Female/MYY/FYY*

Description

This function plots population dynamics overtime showing the count of the individuals separated by sex over the years of the simulation. It is based off the following columns: N_Males_1, N_YYMales_1, N_Females_1 and N_YYFemales of the CDMetaPOP output file summary_popAllTime

Usage

```
pop_sex_plot(data)
```

Arguments

`dataframe`	Import the summary_popAllTime.csv dataframe

Value

a ggplot object representing the overtime population size for the four sexes

pop_year_class *Create abundance by year for each age class*

Description

This function plots population dynamics overtime according by the age of the individuals. It shows the count of the individuals over the years of the simulation separated by age. It is based off the N_Initial_Class column of the CDMetaPOP output file summary_classAllTime

Usage

```
pop_year_class(file_path, n = 5)
```

Arguments

`n`	An integer that specifies the time intervals of the time series. It defaults to 5 years.
`x`	Specify the path to the summary_classAllTime.csv dataframe

Value

a ggplot object representing the general population size at 5 years intervals for the different age classes.

`read.cdmetapop` *Read cdmetapop file as dataframe*

Description

This function imports in r a cdmetapop output file as a dataframe

Usage

```
read.cdmetapop(file_path)
```

Arguments

`x`	Specify the file name and path.

Value

Dataframe

`separate_column` *Column symbol separator*

Description

This function eases the separation of column content generated from cdmetapop outputs. CD-MetaPOP uses delimiters and different delimiter options are specified for different fields. Delimiter '|' or 'bar' is used time parameters. The delimiter '~' ('tilda') is used to assign parameters based on sex. The delimiter ';' is the default and used in various contexts to split fields and patches parameters. The delimiter ':' is used to split parameters used for functions. Note that Loo uses one instance of ';' in an Hindex option. See CDMetaPOP manual for more specific uses and examples.

Usage

```
separate_column(file_path, column_name, sep)
```

Arguments

`x`	Specify the file name and path.

Value

Dataframe of columns for each value that was originally separated by the delimiter.

size_age_class *Function to create a plot of size at age overtime*

Description

This function plots the overtime average size of individuals at a specific age. It is based on the AgeSize_mean columns of the CDMetaPOP output file summary_classAllTime

Usage

```
size_age_class(file_path)
```

Arguments

`x`	Specify the path to the summary_popAllTime.csv dataframe

Value

a ggplot object representing the overtime average sizes of the individuals in the simulation separated by age.

unite_column *Unite multiple columns into one*

Description

A wrapper for tidyr::unite designed specifically to handle large cdmetapop column outputs

Usage

```
unite_column(dataframe, column_name, sep, cols)
```

Arguments

`dataframe`	A data frame
`column_name`	Name of the new column
`sep`	Separator for united columns
`cols`	Columns to unite (character vector of column names)

Value

A data frame with the united columns

Examples

```
# example code
df <- data.frame(
  Subpatch = c("A", "B", "C"),
  N = c(25, 30, 35),
  N_1 = c(34, 44, 34),
  N_2 = c(21, 43, 54)
)

# Specify the columns to unite
cols_to_unite <- c("N_1", "N_2")

# Call the function to unite the columns
result <- unite_column(df, column_name = "N", sep = "|", cols = cols_to_unite)
print(result)

# Example with column names X1 to X100
df2 <- data.frame(matrix(1:10000, ncol = 100))
cols <- names(df2)
result2 <- unite_column(df2, column_name = "All_X", sep = ",", cols = cols)
head(result2)
```

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write_classvars *Build ClassVars File from Template*

Description

This function loads a provided ClassVars template, allows the user to edit selected areas, and saves the modified version as a new file.

Usage

```
write_classvars(output_file = "my_new_classvars.csv")
```

Arguments

`output_file`	The name of the output file. Defaults to 'my_new_classvars.csv'.

Value

A Shiny app instance.

write_patchvars — *Build PatchVars File from Template*

Description

This function loads a provided PatchVars template, allows the user to edit selected areas, and saves the modified version as a new file.

Usage

```
write_patchvars(output_file = "my_new_patchvars.csv")
```

Arguments

`output_file`	The name of the output file. Defaults to 'my_new_patchvars.csv'.

Value

A Shiny app instance.

write_popvars — *Build PopVars File from Template*

Description

This function loads a provided PopVars template, allows the user to edit selected areas, and saves the modified version as a new file.

Usage

```
write_popvars(output_file = "my_new_popvars.csv")
```

Arguments

`output_file`	The name of the output file. Defaults to 'my_new_popvars.csv'.

Value

A Shiny app instance.

`write_runvars` *Build RunVars File from Template*

Description

This function loads a provided runVars template, allows the user to edit selected areas, and saves the modified version as a new file.

Usage

```
write_runvars(output_file = "my_new_runvars.csv")
```

Arguments

`output_file`	The name of the output file. Defaults to 'my_new_runvars.csv'.

Value

A Shiny app instance.

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